Data Analysis: IDL meets Python

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Outline

- Setting up Python-IDL Bridge Environment
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- Calling IDL from Python:
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  - Copying variables from Python to IDL

- Real World Examples:
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  - Calibrate SDO/AIA image
  - Plot GOES light curves
  - Search Virtual Solar Observatory (VSO)

Setting up the Bridge Environment

- Minimum requirements: Python 2.7 and IDL 8.5
- Python 3.4 supported in IDL 8.6
- Anaconda distribution preferred
- Useful links:
  - http://wiki.heliodocs.com/docs (this presentation)
  - https://www.harrisgeospatial.com/docs/pythontoidl.html
  - http://blog.rtwilson.com/ten-little-idl-programs-in-python/
  - http://mathesaurus.sourceforge.net/idl-numpy.html
  - http://sunpy.org

Windows

```
# Set environment variables via Control Panel or use pyidl.bat

IDL_DIR =>
C:\Program Files\Exelis\IDL85

SSW => C:\ssw

SSW_INSTR => aia hmi secchi

PYTHONPATH => where Python modules live
%IDL_DIR%\bin\bin.x86_64;%IDL_DIR%\lib\bridges;%SSW%\gen\python
   \bridge
```

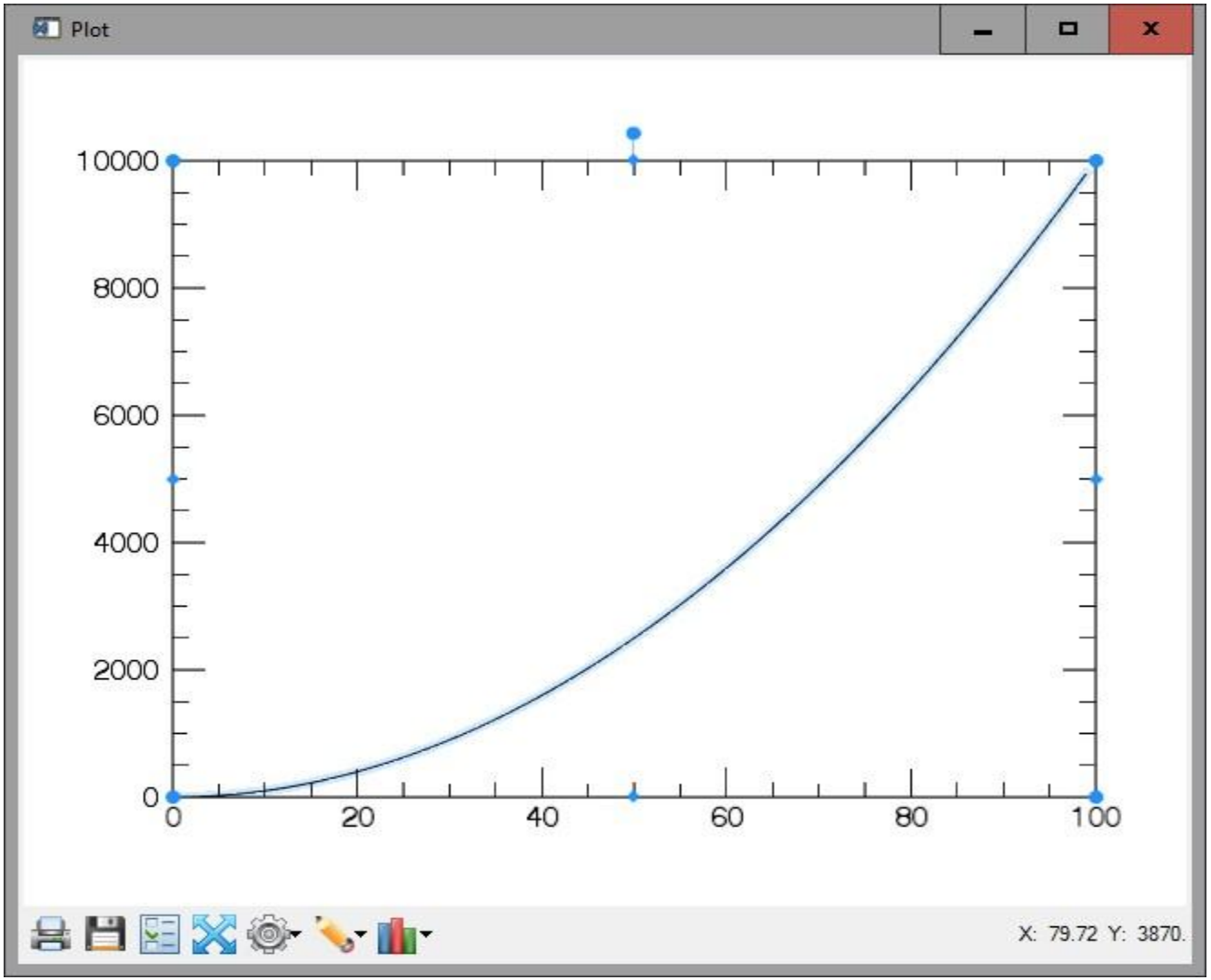
Linux

```
# Set environment variables via .cshrc or .login

setenv IDL_DIR /usr/local/exelis/idl85

setenv SSW /solarsoft

setenv SSW_INSTR “aia hmi secchi”

setenv PYTHONPATH
   $IDL_DIR/bin/bin.linux.x86_64:$IDL_DIR/lib/bridges:$SSW/gen
   /python/bridge

setenv LD_LIBRARY_PATH
   /usr/local/lib/python2.7:$IDL_DIR/bin/bin.linux.x86_64
```

Calling IDL from Python

```
# load Python-IDL bridge module

>>> import bridge

# execute IDL startup within Python

>>> IDL=bridge.setup()

#  use IDL object run method to create and plot array in IDL

>>> IDL.run("x = FINDGEN(100)")
>>> IDL.run("y = x^2")
>>> IDL.run("z=plot(x,y)")
```

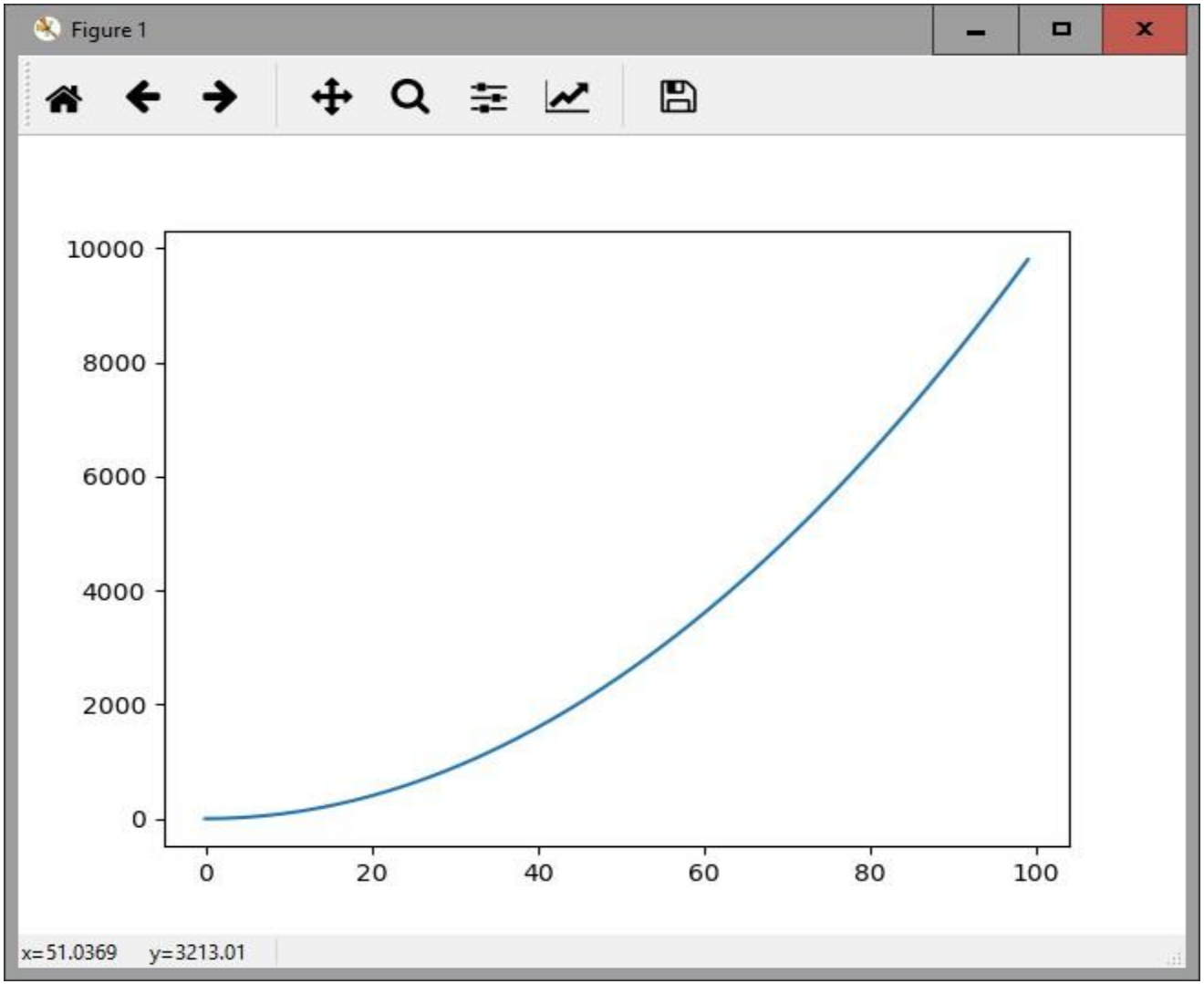
Result in IDL graphics



Copying variables from IDL to Python

```
# copy IDL variables as properties of IDL object

>>> x=IDL.x
>>> y=IDL.y

# IDL arrays become NumPy arrays

>>> x[0:5]
array([ 0.,  1.,  2.,  3.,  4.], dtype=float32)

>>> y[0:5]
array([  0.,   1.,   4.,   9.,  16.], dtype=float32)

# plot NumPy arrays using Matplot Pyplot

>>> import matplotlib.pyplot as plt
>>> plt.plot(x,y)
>>> plt.show()
```

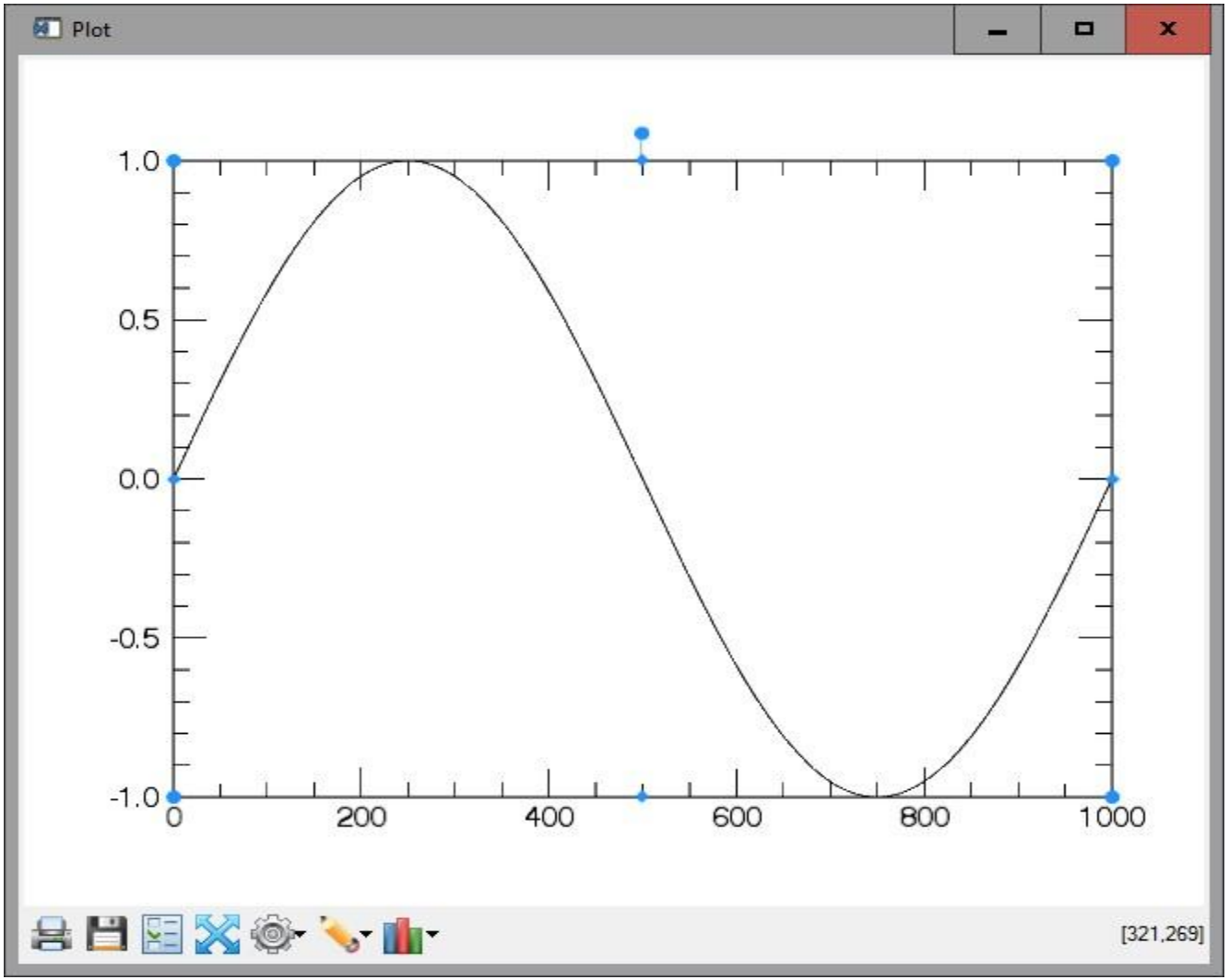
Result in Python graphics



Copying variables from Python to IDL

```
# create NumPy arrays

>>> x=np.arange(1000)
>>> y=np.sin(2*np.pi*x/1000)

# copy arrays as properties to IDL object

>>> IDL.x=x
>>> IDL.y=y

# call IDL plot command as a method

>>> IDL.plot(x,y)
```

Result in IDL graphics



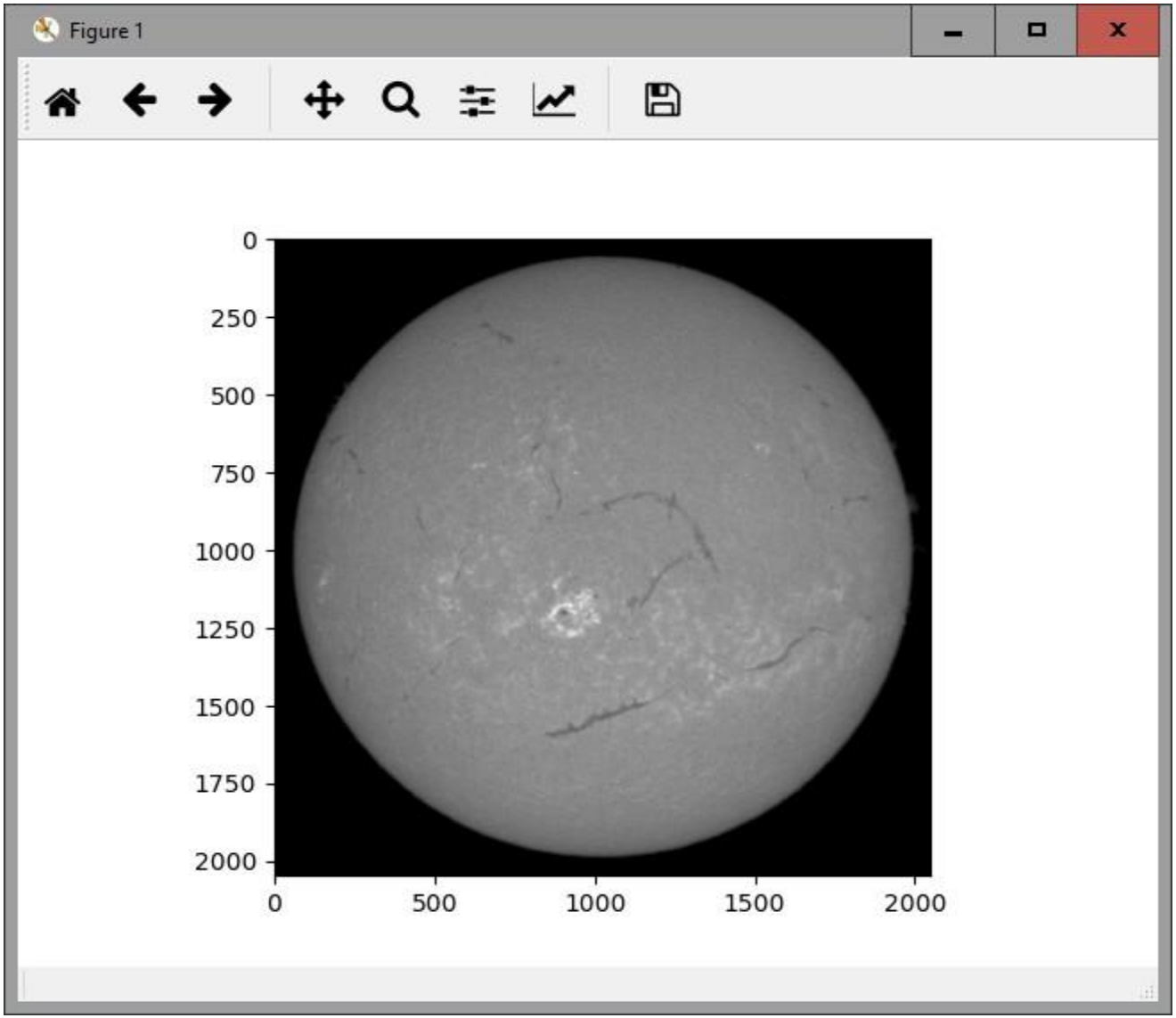
Read FITS file

```
# call IDL FITS reader from Python

>>> IDL.file="bbso_halph_fl_20150312_185727.fts"
>>> IDL.run("data=readfits(file,header)")

# copy FITS header Python

>>> header=IDL.header
>>> type(header)
<type ‘list’>

>>> header[0:4]
['SIMPLE  =                    T / Written by IDL:  Thu Mar
   12 12:38:26 2015      ', 'BITPIX  =                   16 /
   ', 'NAXIS   =                    2 /
   ', 'NAXIS1  =                 2048 /
   ']
```

Display FITS image

```
# copy FITS data to Python

>>> data=IDL.data
>>> type(data)
<type 'numpy.ndarray'>

# flip image North-South

>>> import matplotlib.pyplot as plt
>>> import numpy as np
>>> plt.imshow(np.flipud(data),cmap='grey')
>>> plt.show()
```

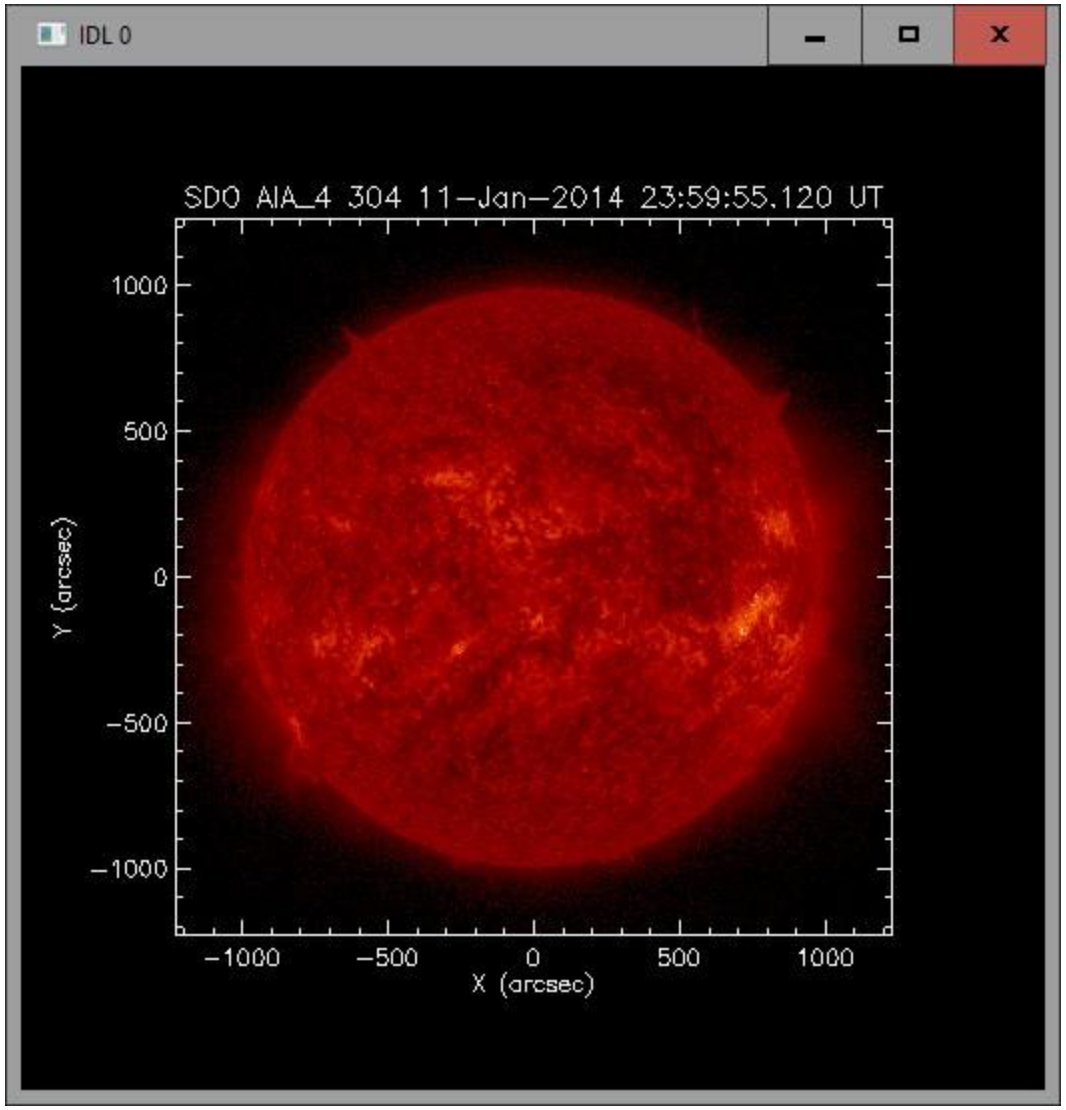
Result in Python graphics



Read and prep SDO/AIA image

```
# Insert IDL commands into a batch file (e.g. script.pro)

 file='aia.lev1.304A_2014-01-11T23_59_55.12Z.image_lev1.fits'
 read_sdo,file,index,data
 aia_prep,index,data,pindex,pdata
 index2map,pindex,pdata,amap
 aia_lct,wave=304,/load
 plot_map,amap,/log

# Execute batch file from Python

>>> IDL.run("@script")
```

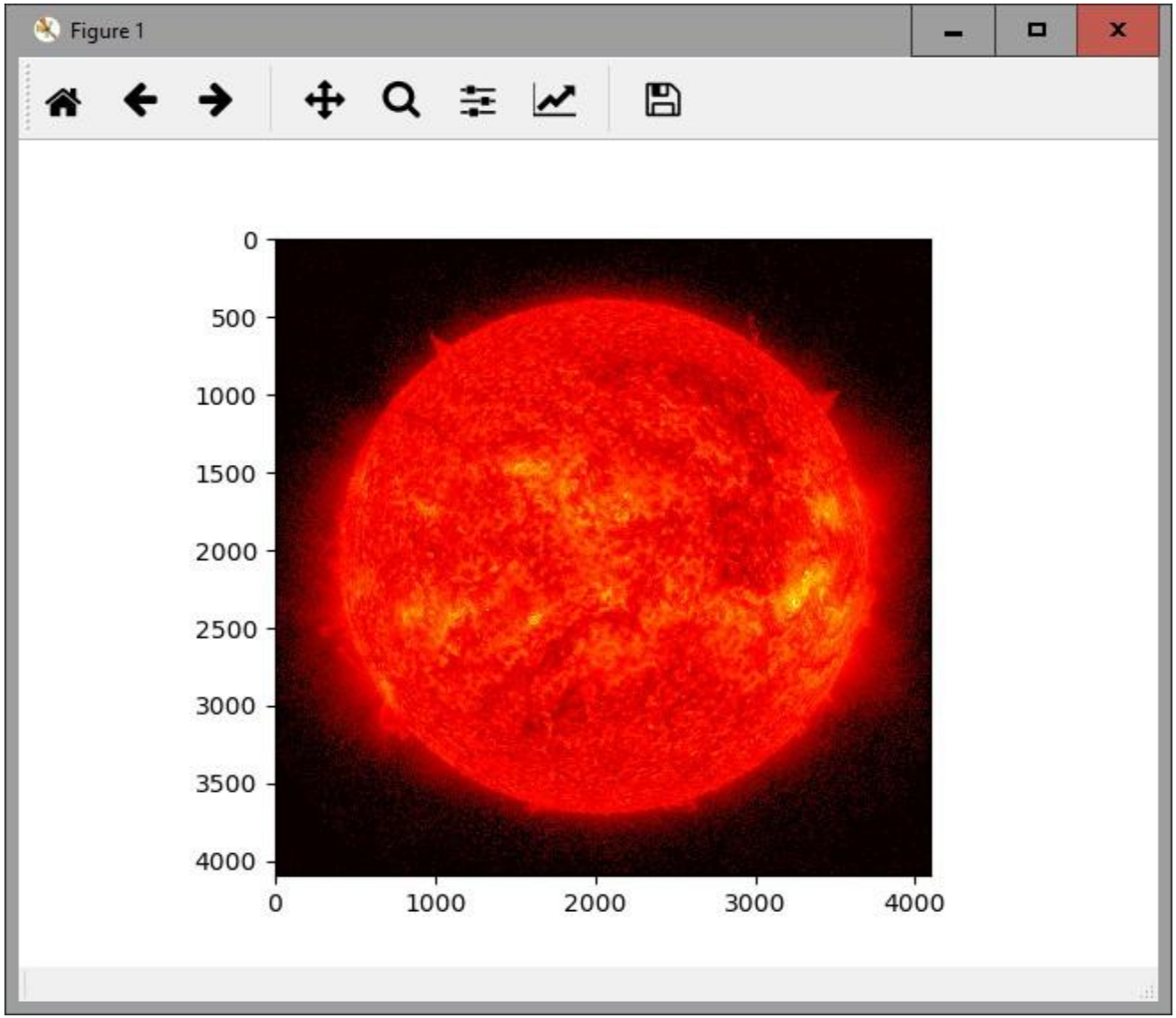
Result in IDL graphics



Extract SDO/AIA image map

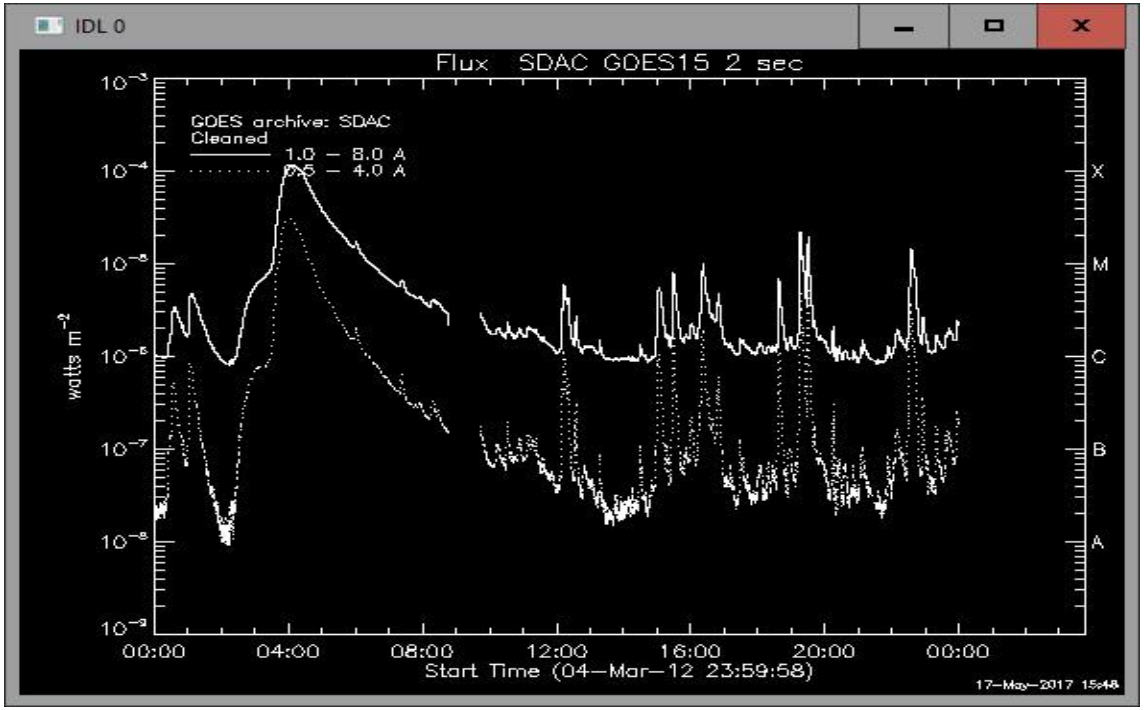
```
# copy IDL map structure to Python dictionary

>>> map=IDL.amap
>>> type(map)
<type 'dict'>

>>> map.keys()
['XUNITS', 'RSUN', 'L0', 'ROLL_ANGLE', 'XC', 'YC', 'DY',
   'ROLL_CENTER', 'YUNITS', 'SOHO', 'DX', 'TIME', 'B0', 'DUR',
   'DATA', 'ID']

# plot on log scale (flip and remove zeroes)

>>> data=np.flipud(map['DATA'])
>>> (min, max) = (data[data > 0].min(), data.max())
>>> ldata = np.log10(data.clip(min,max))
>>> plt.imshow(ldata,cmap='hot')
>>> plt.show()
```

Result in Python graphics



Read and plot GOES lightcurves

```
# create IDL GOES object

>>> IDL.run("goes=ogoes()")
>>> IDL.run("goes->read,'5-mar-12','6-mar-12'")
>>> IDL.run("goes->plot")
```

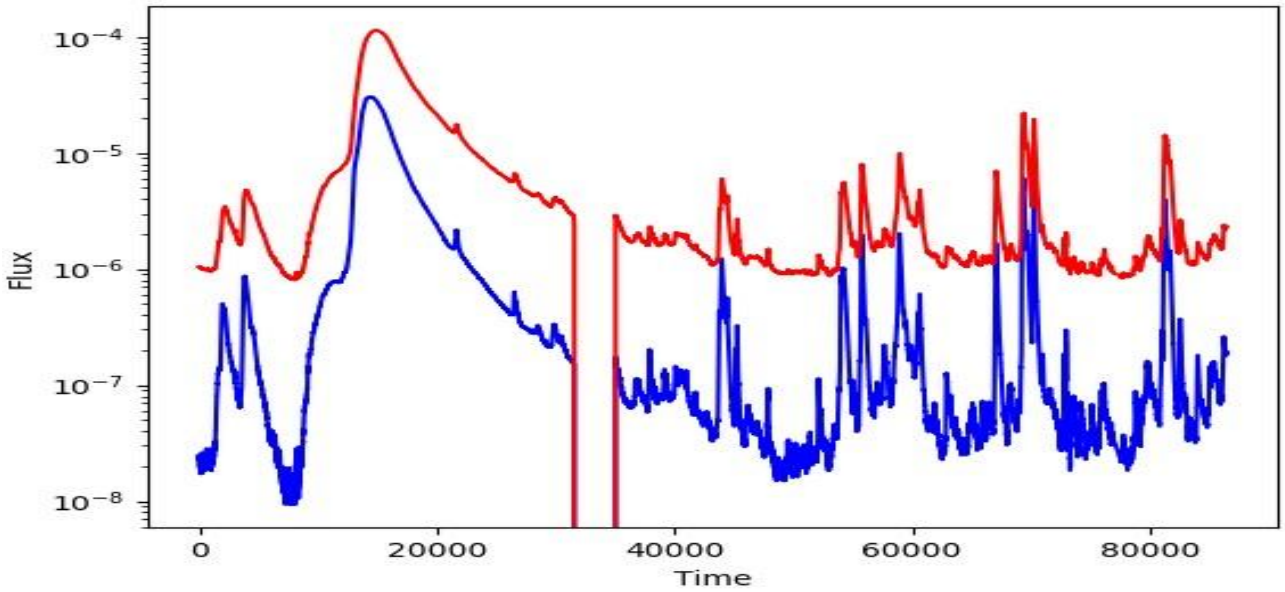


Copy GOES lightcurve data into Python

```
#  copy IDL data array into NumPy array

>>> flux=IDL.goes.data
>>> flux
array([[  1.05180004e-06,   1.05699996e-06,   1.05439995e-06, ...,
          2.32139996e-06,   2.32139996e-06,   2.32139996e-06],
       [  2.38909994e-08,   2.27769998e-08,   2.50060008e-08, ...,
          1.89950001e-07,   1.88830001e-07,   1.88830001e-07]],
  dtype=float32)

>>> time=IDL.goes.time
>>> time
array([  0.00000000e+00,   2.05000007e+00,   4.09600008e+00, ...,
         8.63958000e+04,   8.63978500e+04,   8.63998960e+04])
```

Plot GOES lightcurves in Python

```
>>> import matplotlib.pyplot as plt
>>> plt.plot(time,flux[0,:],color='r')
>>> plt.plot(time,flux[1,:],color='b')
>>> plt.yscale('log')
>>> plt.xlabel('Time')
>>> plt.ylabel('Flux')
>>> plt.show()
```



Search VSO from Python

```
 # copy search command for STEREO/EUVI from Python to IDL

>>> cmd="files=vso_search('1-apr-17','2-apr-17',inst='euvi',/url)"
>>> IDL.cmd=cmd
>>> IDL.run(cmd)

# copy IDL results back to Python as List

>>> files=IDL.files

# each list element is a Dictionary

>>> file[0].keys()
['INFO', 'SOURCE', 'URL', 'PROVIDER', 'ECLIPSE', 'PERCENTD', 'WAVE',
   'DARK', 'INSTRUMENT', 'GETINFO', 'EXTENT', 'TIME', 'PHYSOBS',
   'DETECTOR', 'SIZE', 'EXPTIME', 'FILEID']

>>> file[0]['URL']
'https://stereo-
   ssc.nascom.nasa.gov/data/ins_data/secchi/L0/a/img/euvi/20170401//
   20170401_000530_n4euA.fts'
```

python + IDL

Works great!